

- **Search Archives:** Many questions have already been answered.

Tips for community engagement

- **Be Specific:** When asking questions, provide code snippets and error messages.
- **Contribute:** Help others when you can to reinforce your own learning.

2) Learn data literacy skills

Data literacy is the ability to read, understand, and communicate data as information. In a data-driven world, these skills are invaluable.

What exactly is data literacy

Importance of data in today's digital world

- **Informed Decision-Making:** Data helps businesses and individuals make evidence-based decisions.
- **Competitive Advantage:** Organisations leverage data analytics for market insights and trends.
- **Personal Empowerment:** Understanding data allows you to interpret statistics and reports critically.

Types of data skills: Collection, analysis, and visualisation

- **Data Collection:** Gathering data through surveys, sensors, or web scraping.
- **Data Analysis:** Using statistical methods to interpret data sets.
- **Data Visualization:** Presenting data in graphical formats to communicate findings effectively.

Basic tools for data analysis

Learning Microsoft Excel for data management

- **Why Excel?:** Widely used and accessible tool for data manipulation.
- **Key Features:**
 - **Formulas and Functions:** Perform calculations and data analysis.
 - **PivotTables:** Summarise large data sets.
 - **Data Visualization:** Create charts and graphs.

How to get started with Excel:

1. **Input Data:** Organise data in rows and columns.
2. **Use Formulas:** Apply functions like SUM, AVERAGE, and VLOOKUP.
3. **Create Charts:** Highlight data and insert charts to visualise.

Introduction to Google Data Studio and Tableau for visualisation

- **Google Data Studio:**
 - **Free Tool:** Create interactive dashboards.
 - **Data Sources:** Connect to Google Sheets, Google Analytics, and more.

- **Sharing:** Easily share reports with others.

■ **Tableau:**

- **Powerful Visualisation:** Advanced features for complex data.
- **User-Friendly Interface:** Drag-and-drop functionality.
- **Community Support:** Extensive online resources and forums.

How to use these tools effectively:

1. **Connect Data Sources:** Import your data into the platform.
2. **Choose Visualisation Types:** Select from charts, maps, or custom visuals.
3. **Customise Dashboards:** Tailor the layout and design.

R and Python for advanced data analysis

■ **R:**

- **Statistical Computing:** Designed for data analysis and visualisation.
- **Packages:** Utilise libraries like ggplot2 for plotting.

■ **Python:**

- **Data Libraries:** Use pandas for data manipulation, matplotlib, and seaborn for visualisation.
- **Versatility:** Combine data analysis with other programming tasks.

Learning resources you can access:

- **Online Courses:** Platforms like DataCamp and Coursera offer courses in R and Python.
- **Documentation:** Refer to official documentation for in-depth understanding.

How to build simple data dashboards

Creating dashboards helps in monitoring key metrics at a glance. Here's how you can get started on your first data visualisation project.

Using Google Sheets for basic dashboards

■ **Step-by-Step Guide:**

- **Organise Data:** Ensure data is clean and structured.
- **Create Charts:** Use the Insert Chart function.
- **Design Dashboard:** Arrange charts and tables in a separate sheet.

■ **Benefits:**

- **Accessibility:** Shareable and editable in real-time.
- **Cost-Effective:** Free tool with robust features.

Customising Charts and Graphs to Tell a Story

■ **Choose the Right Chart Type:**

- **Line Charts:** Show trends over time.
- **Bar Charts:** Compare quantities.
- **Pie Charts:** Display proportions.

■ **Best Practices:**

- **Keep It Simple:** Avoid clutter.
- **Use Colours Wisely:** Highlight key data points.
- **Add Labels and Titles:** Provide context.